Conformal Uncertainty Quantification for Mammographic Malignancy Classification

Knowing When the Model Doesn't Know

Authors: Darshan S Gowda, Rakshith K B

Course: MSc Data & Computational Science

Year: 2026-27

Supervisor: Dr. Sarp Akçay **Dataset:** CBIS-DDSM — 3,568 biopsy-confirmed mammograms **Method:** Split conformal prediction (LAC / APS / RAPS + Mondrian) **Coverage target:** α = 0.05 → 95 % guarantee

A DenseNet-121 mammography classifier is wrapped with split conformal prediction — turning uncalibrated scores into label sets guaranteed to contain the true diagnosis 95 % of the time, automatically flagging ambiguous cases for radiologist review.

1 The Problem

Our classifier's raw confidence is off by **ECE = 0.205** — a 20-point gap between stated confidence and actual accuracy. This overconfidence is a known property of softmax networks, not a flaw in this specific model.

What screening actually needs: predictions with a *stated, guaranteed error rate* — not a number that merely looks like a probability. A confidently wrong "benign" is a missed cancer.

THE COVERAGE GUARANTEE

$\Pr(Y_{true} \in C(X)) \geq 1 - \alpha \quad (\alpha = 0.05 \rightarrow 95\,\%\text{ coverage})$

When the set holds **one label** the model is confident and the case can be auto-reported. When it holds **both labels** the model is honestly flagging ambiguity — that case routes to a radiologist.

2 Approach & Pipeline

Backbone: DenseNet-121 pretrained on ImageNet, fine-tuned on CBIS-DDSM for binary (benign / malignant) classification. All splits are at the **patient level** to prevent data leakage and preserve the conformal guarantee.

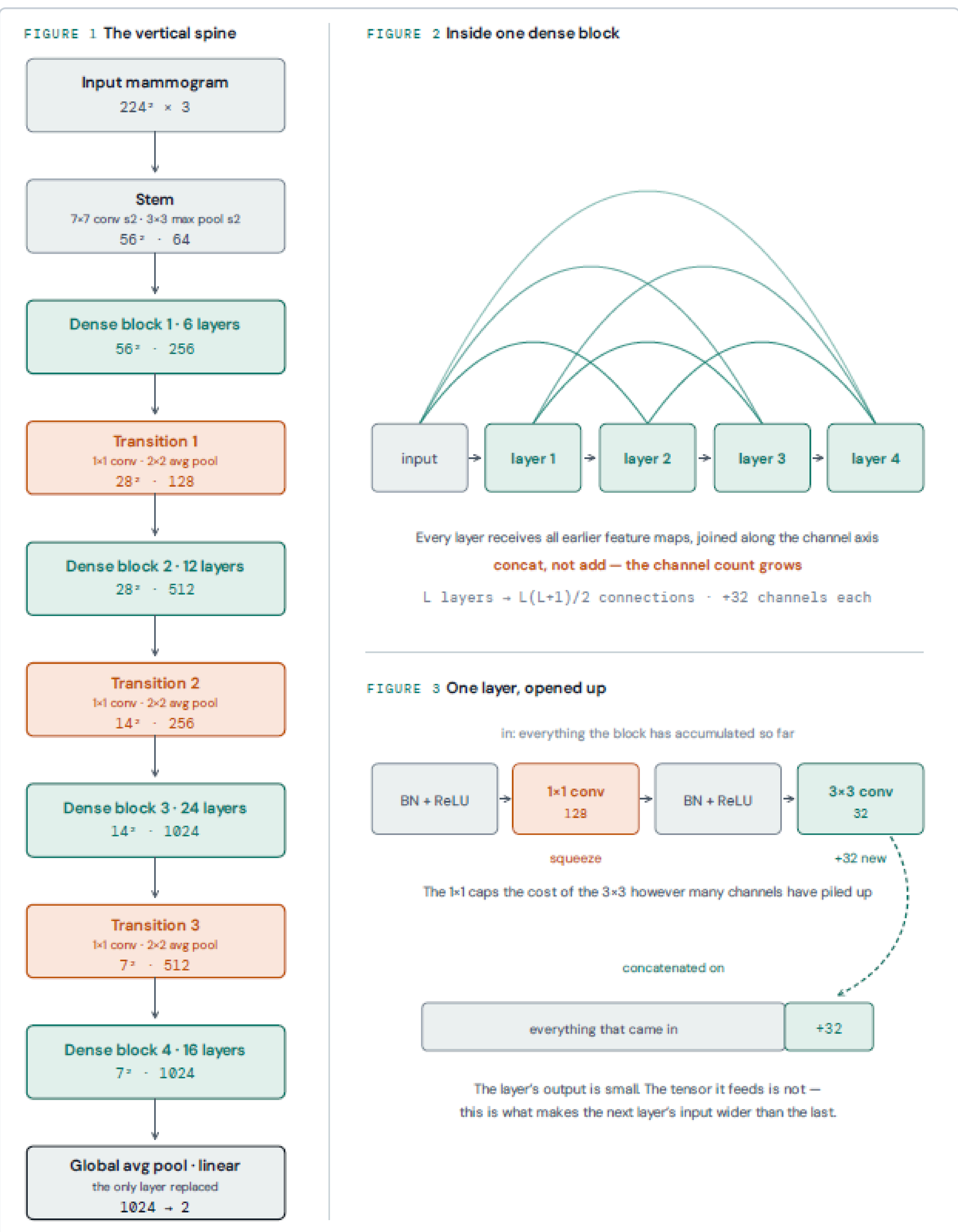
Train 1,951 · Val 352 · Calibration 595 · Test 670

The **calibration split** is the element unique to conformal prediction: never seen during training, used solely to compute the threshold $\hat{q}$ that delivers the coverage guarantee.

THRESHOLD & SET CONSTRUCTION

$\hat{q} = \lceil (n+1)(1-\alpha) \rceil / n \quad \text{quantile of} \quad \{1 - p_{true,i}\} \quad \text{on cal set}$

$C(x) = \{\, k \;:\; p_k(x) \geq 1 - \hat{q} \,\}$



Dense 121: Process Flow

3 The Three Non-Conformity Scores

Three scoring rules are compared. They share the same calibration framework but differ in how the non-conformity score is defined, leading to different set sizes.

Method	Score definition	Set size ± sd	Singleton %
LAC	$1 - p_{true}$	1.694 ± 0.027	30.6 %
APS	Cumulative sorted mass	1.720 ± 0.028	28.0 %
RAPS	APS + set-size penalty	1.714 ± 0.029	28.6 %

All three hold ≥ 95 % coverage. **LAC gives the tightest sets on a binary task** — consistent with theory that RAPS and APS earn their added complexity only when the label space is large. Comparing all three, rather than asserting one, is itself a methodological contribution.

4 Why Our Numbers Are Trustworthy

Before touching any mammogram, all three methods were validated on **synthetic data with known statistics** across 200 independent trials at K = 2 and K = 5. This separates a *correct implementation* from a *fortunate result*.

LAC K=2 0.951 ✓ · APS K=2 0.955 ✓ · RAPS K=2 0.953 ✓

References: Angelopoulos & Bates (2021) · Romano, Sesia & Candès (2020) · Angelopoulos et al. (2021, RAPS) · Sadinle et al. (2019, LAC) · Sawyer-Lee et al. (2016, CBIS-DDSM)

5 Results

Base classifier (12 epochs, single-view): val accuracy 0.733 · test accuracy 0.643 · **AUC 0.708**. Single-view CBIS-DDSM is genuinely hard; the uncertainty layer is the contribution.

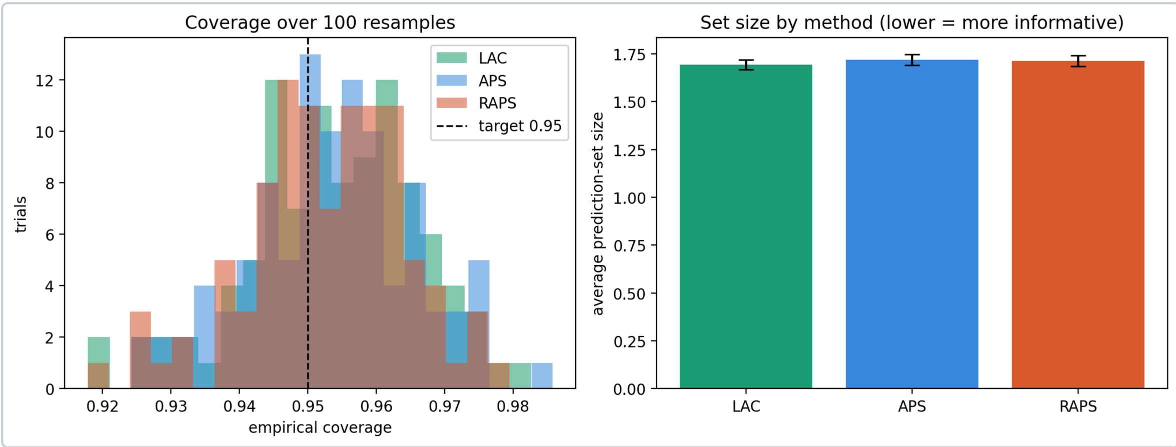


Figure 1. *Coverage distributions and average set sizes across 100 resamples.*

Left: all three methods cluster tightly around the 0.95 target (dashed) across 100 independent calibration/test resamples — the guarantee holds, not just on one lucky split. **Right:** LAC (teal) achieves the smallest average set size (1.694), resolving the most cases to a single confident label. This is the expected result for a binary task and validates the method selection.

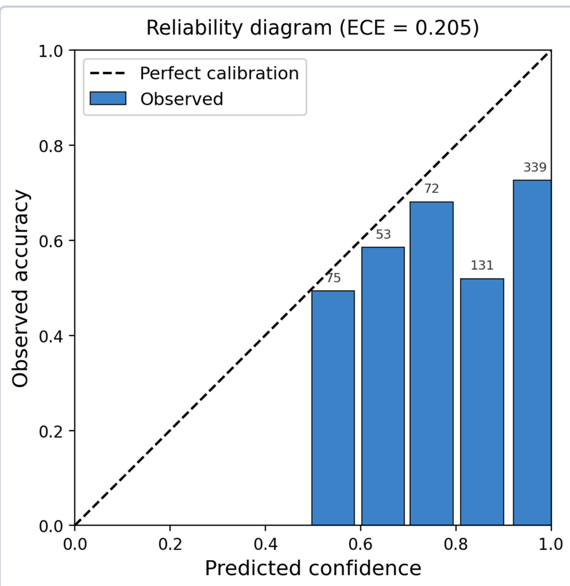


Figure 2. *Reliability diagram (ECE = 0.205).*

Bars below the diagonal = over-confident model. ECE = 0.205 quantifies the 20-point gap between stated confidence and actual accuracy. This is the concrete motivation for conformal prediction.

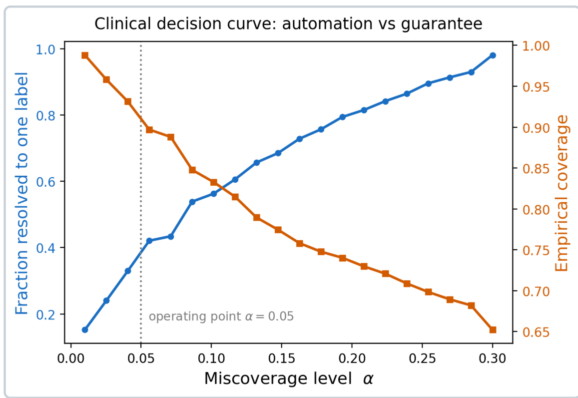


Figure 3. *Decision curve: singleton rate vs α.*

As α rises, more cases resolve to one label (blue) but coverage falls (orange). At α = 0.05: 42 % auto-resolved at 99 % coverage. This is the operating-point curve a clinical team would use.

6 Real Mammograms with Prediction Sets

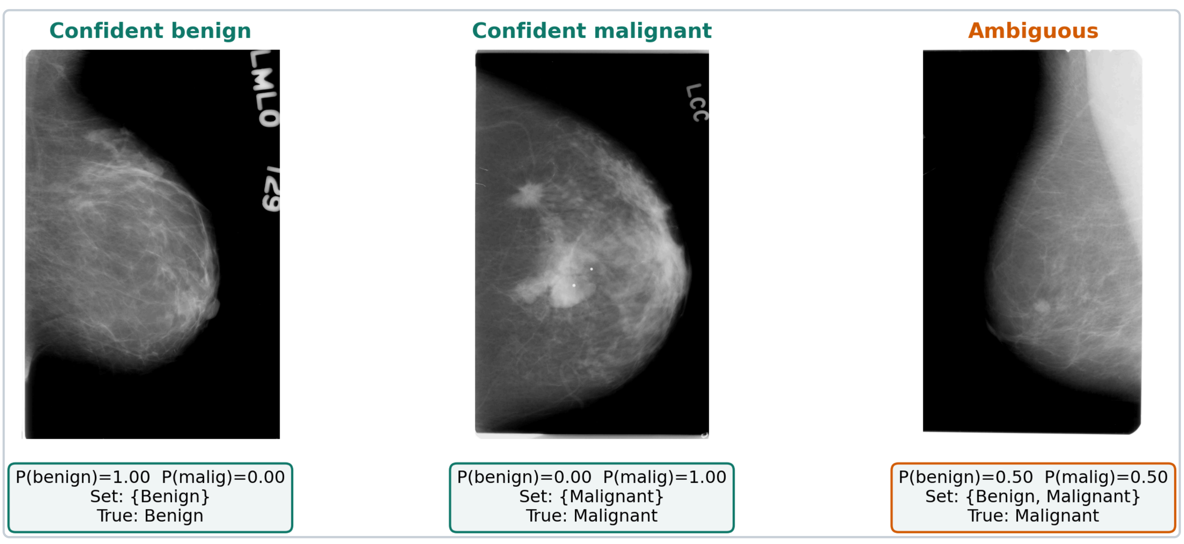


Figure 4. *Three test-set mammograms with LAC prediction sets (α = 0.05).*

Three real cases from the held-out test set show the system working in practice. **Left (Confident benign):** P(malignant) = 0.00 — the set collapses to {Benign} and the case can be auto-reported without radiologist input. **Centre (Confident malignant):** P(malignant) = 1.00 — the set is {Malignant} with full confidence; the case is flagged for immediate follow-up. **Right (Ambiguous):** P(benign) = P(malignant) = 0.50 — both labels are included; the system honestly flags its uncertainty. The true label is Malignant: this is exactly the case that must not be auto-reported, and the prediction set correctly prevents that.

The **set size is the uncertainty signal**: a singleton means confidence; a two-label set means honest admission of ambiguity and automatic escalation to expert review.

7 Class-Conditional Coverage: A Fix

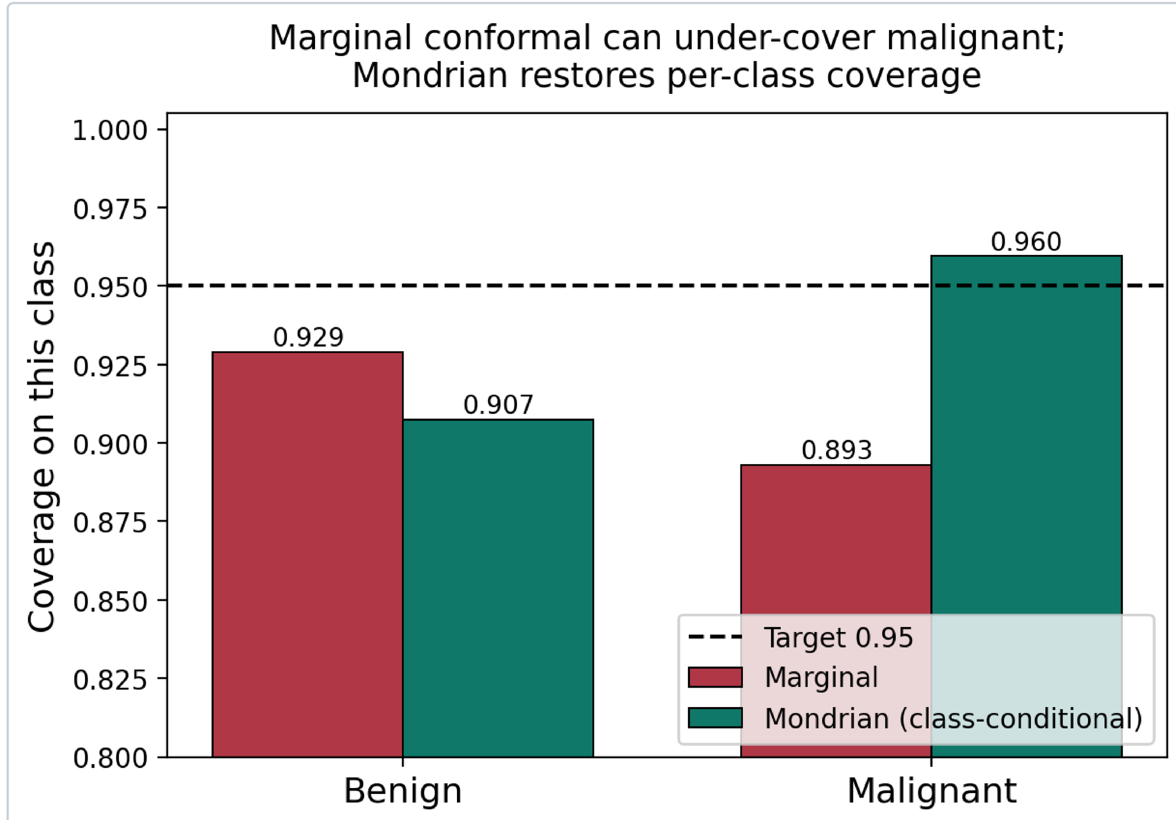


Figure 5. *Marginal vs Mondrian per-class coverage.*

Red bars (Marginal): standard split conformal holds 95 % overall but under-covers malignant at 0.893, meaning 11 % of true cancers fall outside the prediction set — the most dangerous failure mode in screening. **Teal bars (Mondrian):** calibrating a separate threshold per class restores malignant coverage to 0.960 by construction. Mondrian conformal achieves per-class guarantees at the cost of using only the per-class calibration count; with 595 calibration points the estimate is stable.

Key clinical finding: marginal conformal is insufficient in imbalanced screening settings. Mondrian conformal is the correct tool when per-class safety guarantees are required — and it is the tool that should be deployed.

8 Conclusions

- **The guarantee holds empirically.** All three methods achieve ≥ 95 % coverage across 100 resamples, verified on synthetic data before any real data was used — a correct implementation, not a fortunate result.
- **LAC is most efficient on a binary task** (set size 1.694 vs 1.720/1.714), reproducing the theoretical prediction that RAPS earns its complexity only with large label spaces.
- **Marginal conformal under-covers malignant (0.893).** Mondrian per-class calibration restores it to 0.960. This is a clinical distinction, not an academic one: 11 % of missed cancers is unacceptable in screening.
- **ECE 0.205 provides concrete evidence** that raw softmax confidence cannot be trusted, motivating the uncertainty layer with a measurable fact.
- **The decision curve** turns α from a technical knob into a quantified operating-point choice for clinical teams.

9 Future Work

- **Multi-view bilateral fusion.** The Siamese DenseNet-121 architecture with gated cross-view attention is designed and implemented; the conformal layer transfers unchanged to its outputs.
- **BI-RADS multi-class grading.** The conformal code already handles K > 2 (verified at K = 5); extending to five-class BI-RADS is a one-line label change.
- **Cross-dataset generalisation.** Deployment on VinDr-Mammo requires weighted or Mondrian conformal variants to handle distribution shift while preserving the per-class coverage guarantee.
- **Temperature scaling.** Reducing the base classifier's ECE (currently 0.205) before applying conformal prediction would tighten prediction sets further while keeping the guarantee intact.

10 Summary & Code Repository

"We take a breast-cancer image classifier and wrap it in a mathematical procedure that turns its overconfident guesses into honest, guaranteed statements of uncertainty — so the system knows when to defer to a radiologist."

The full pipeline — data loading, DenseNet-121 training, conformal calibration (LAC / APS / RAPS), Mondrian fix, and all four extended analyses — is available as a reproducible Kaggle notebook. The standalone `conformal.py` module and synthetic-data verification script are included so the guarantee can be checked independently of any model.

Scan for the full repository

Dataset: CBIS-DDSM (awsaf49/Kaggle) · Backbone: DenseNet-121 (torchvision) · Conformal: split LAC / APS / RAPS + Mondrian